

Meander-Line Polarizers for Oblique Incidence

C.1 INTRODUCTION

A polarizer is a device that transforms a linear polarized wave into a circular polarized wave, or vice versa. The common principle is simply to decompose the incident field into two components where the phase of one is advanced and the other is delayed such that their difference is 90° while their amplitudes are the same. It appears that Pakan [128] was the first to utilize this principle. Later improvements were introduced by Lerner [129]. These devices were not of the meander-line type, as will be discussed here. These seem to appear first in a paper by Young et al. [130] and were subsequently improved by Epis [131]. Later, a paper by Terret et al. [132] discussed how to calculate the susceptance of a meander line. All of these contributions were primarily focused on normal angle of incidence while Chu and Lee [133] extended the calculation to include oblique angle of incidence. A recent contribution was supplied by Marino [134]. It was apparent that meander-line polarizers gradually deteriorate for higher angles of incidence. The present appendix will demonstrate that introduction of a “dielectric profile” can greatly improve this calamity.

The general principle for a meander-line polarizer is illustrated in Fig. C.1, top. It consists of one (or more) sheet(s) of meander-line-shaped conductors usually printed on a circuit board. The E vector $\vec{E}^i$ of an incident plane wave is tilted 45° with respect to the principal axis of the meander line. It is decomposed into a vertical component $\vec{E}_v^i$ and a horizontal component $\vec{E}_h^i$. The equivalent circuit for the vertical component is a transmission line with a shunt inductance

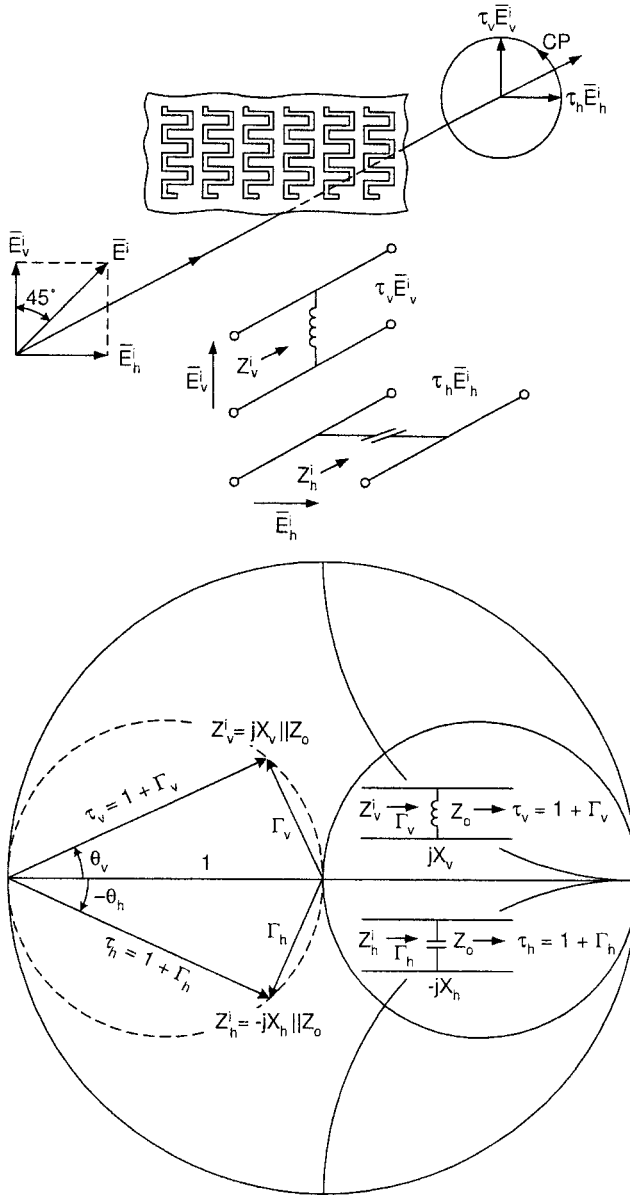


Fig. C.1 The workings of a single meander-line polarizer. Top: A meander-line sheet acts as a shunt inductance for the vertical components and as a shunt capacitance for the horizontal components of an incident signal. Bottom: The vertical component of the incident field is reflected with reflection coefficient Γ_v and transmitted with the transmission coefficient $\tau_v = 1 + \Gamma_v$ as shown in the Smith chart. Similarly, the horizontal component is reflected and transmitted with reflection coefficient Γ_h and transmission coefficient $\tau_h = 1 + \Gamma_h$, respectively.

as illustrated in Fig. C.1, middle, while the equivalent circuit for the horizontal component is a transmission line with a shunt capacitor, as also illustrated in the same figure.

Let us denote the reactance of the equivalent inductance by jX_v and that of the equivalent capacitance by $-jX_h$ as indicated in the insert of the Smith chart, Fig. C.1, bottom. Then the input impedances for the two components are going to be $Z_v^i = jX_v \parallel Z_0$ and $Z_h^i = -jX_h \parallel Z_0$, respectively. These two values are also indicated in the Smith chart, and we note that they are located on the great circle passing through (0, 0) and the center of the Smith chart (for details consult Appendix B). We now further remind the reader that one of the many beautiful features of the Smith chart is that the distance from the center to the respective input impedances located in the Smith chart simply constitute the reflection coefficients Γ_v and Γ_h , respectively, as also indicated in the same Smith chart (the phase is measured from the horizontal line going right). Finally, the transmission coefficients $\tau_v = 1 + \Gamma_v$ and $\tau_h = 1 + \Gamma_h$ are obtained by simple vector addition of the respective reflection coefficients and “1,” namely, the radius of the Smith chart as also indicated. Again, the phase angles θ_v and $-\theta_h$ of the respective transmission coefficients are measured from the horizontal line. The total phase difference between these two transmitted components is seen to be $\theta_v + \theta_h$. When the difference is 90° and if $|\tau_v| = |\tau_h|$, the total transmitted field will be circular polarized.

There will also be a reflected field. The vertical and horizontal components have amplitudes and phases given simply by Γ_v and Γ_h , respectively. In general this reflected field will be elliptic polarized. What is of concern, however, is the fact that when a single meander-line sheet transmits a perfect circular polarized field, we readily see from the Smith chart that $|\Gamma_v| = |\tau_v| = |\Gamma_h| = |\tau_h|$. In other words: A single meander-line sheet will produce a reflected field of the same amplitude as the transmitted field; that is, the efficiency is only 50%. The remedy for this calamity is to use several meander-line sheets cascaded after each other as will be discussed in the next section. See also Problem C.1, where you are asked to consider a two-sheet meander-line polarizer.

C.2 MULTILAYERED MEANDER-LINE POLARIZERS

Use of two or more meander-line sheets can, if designed correctly, result in much improved performance. An example consisting of three sheets is shown in Fig. C.2, middle, where we show the equivalent circuit for the vertical component only. We shall start our analysis from the back, where the impedance at point 1 is equal to $jX_3 \parallel Z_0$ as indicated in the Smith chart at the top. This point is now rotated to point 2, corresponding to the sheet separation d_3 . We next add the sheet impedance jX_2 in parallel and arrive at point 3. Note that in order to preserve “symmetry” we choose $jX_2 \sim \frac{1}{2}jX_3$. This will bring point 3 to a position similar to point 1 (in general all “inside” sheets should have a susceptance twice as large as the two “outside” sheets). This simply means that the transmission angle θ_{v2}

where $\theta_{v2} \sim 2\theta_{v1} \sim 2\theta_{v3}$, that is,

$$\theta_{v1} = \theta_{v3} = \frac{45^\circ}{4} = 11, 25^\circ,$$

$$\theta_{v2} = 22.5^\circ.$$

Note: In addition to the phase advance of 45° for the vertical component, there will also be a delay from the sheet separations d_2 and d_3 . However, this delay will also be experienced by the horizontal component (that is, also delayed 45° due to the capacitive effect). Thus, the net phase difference between the two components is still going to be 90° as desired. In the following we shall show various designs. They differ essentially by the distance between the sheets as well as by the dielectric constant of the spaces that typically ranges from air to $\epsilon_r \sim 2$.

C.3 INDIVIDUAL MEANDER-LINE IMPEDANCES

The impedance of the outer meander-line sheet including the free space Z_0 behind the individual sheets is shown in the Smith chart in Fig. C.3 for the vertical as well as the horizontal components. Similarly, we show the impedances for the inner meander-line sheets including Z_0 in the Smith chart shown in Fig. C.4. We observe that both the vertical and horizontal components vary with frequency; however, the angular difference between the transmission coefficients for the two components remains reasonably constant with frequency. This feature is quite essential for producing polarizer designs with a large frequency range.

We also note that the angles of the transmission coefficient for the inner meander lines are approximately twice that for the outer meander lines, as they should be (see discussion above). The meander-line dimensions are given in the respective Smith charts. We observe that both of these designs have the same dimensions D_x and D_z , and so on. This is done so as not to violate Floquet's Theorem (see reference 26). The differences in impedances are obtained by using different line widths w in the two cases.

We further note that all the meander lines are encapsulated by dielectric sheets ("underwear") of thickness 2×20 mils and relative dielectric constant $\epsilon_r = 2.2$.

To assist in designing the proper meander-line sheets, we show in Fig. C.5 the typical impedance variance as a function of the strip width w and meander-line spacings D_x . Note that the variation is one-sided as a function of the strip width w and symmetric with meander-line spacing D_x .

C.4 DESIGN 1

Meander-Line Sheet Separation: $\sim \lambda_0/4$

Dielectric: Air

Para-Para Pol. (Solid) / Perp-Perp Pol. (Dashed)
 Z_0 (Para-Para) = 376.73 Ohms
 Z_0 (Perp-Perp) = 376.73 Ohms

Frequency Range: 6.0 GHz - 9.9 GHz
 $\alpha = 0.0$ Deg. and $\epsilon = 0.0$ Deg.

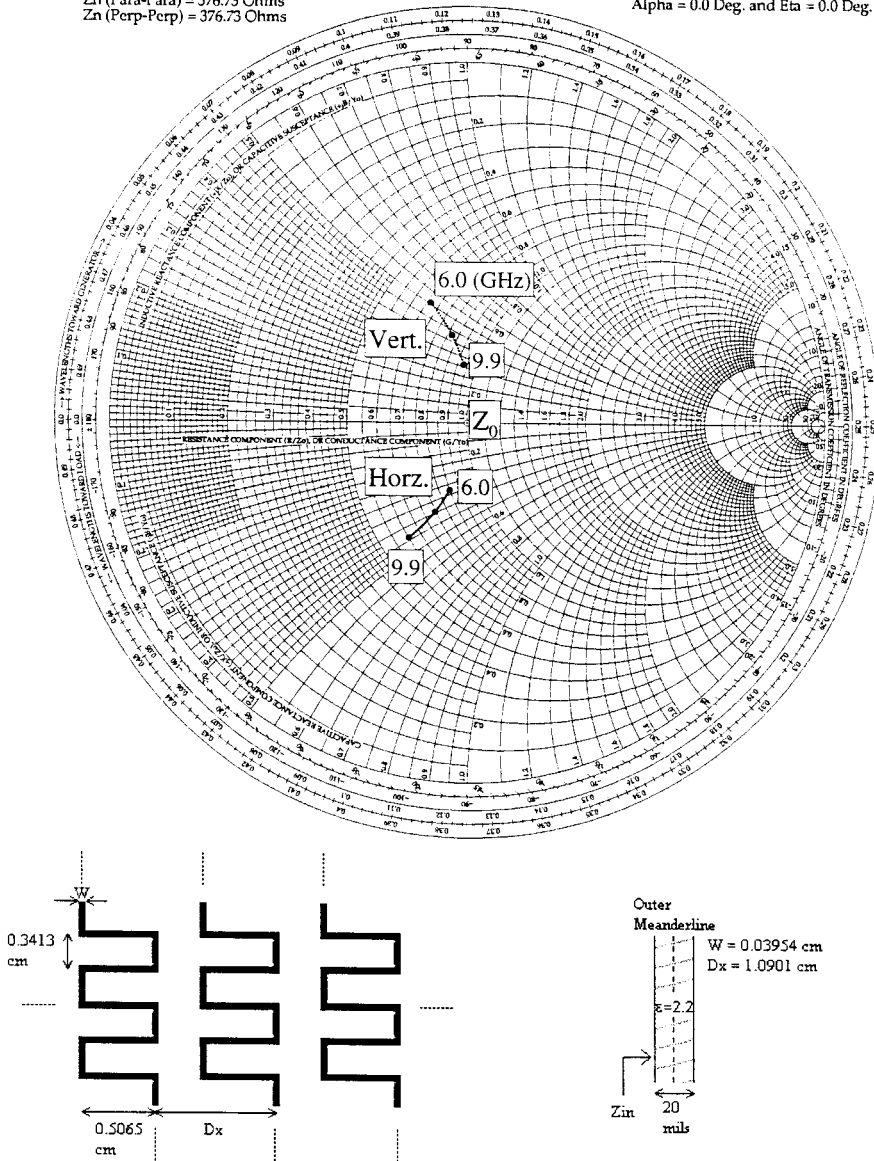


Fig. C.3 The input impedance of a single encapsulated meander line (outer) including free space Z_0 behind. Vertical as well as horizontal cases.

This design is comprised of three meander-line sheets separated by air of thickness 0.8657 cm, as shown at the bottom of Fig. C.6. At $f = 8.7$ GHz this corresponds to a separation of $\sim \lambda_0/4$. However, when incorporating the dielectric substrate around the meander lines, an effective quarter-wave separation is obtained at

Para-Para Pol. (Solid) / Perp-Perp Pol. (Dashed)
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Frequency Range: 6.0 GHz - 9.9 GHz
 $\alpha = 0.0$ Deg. and $\eta = 0.0$ Deg.

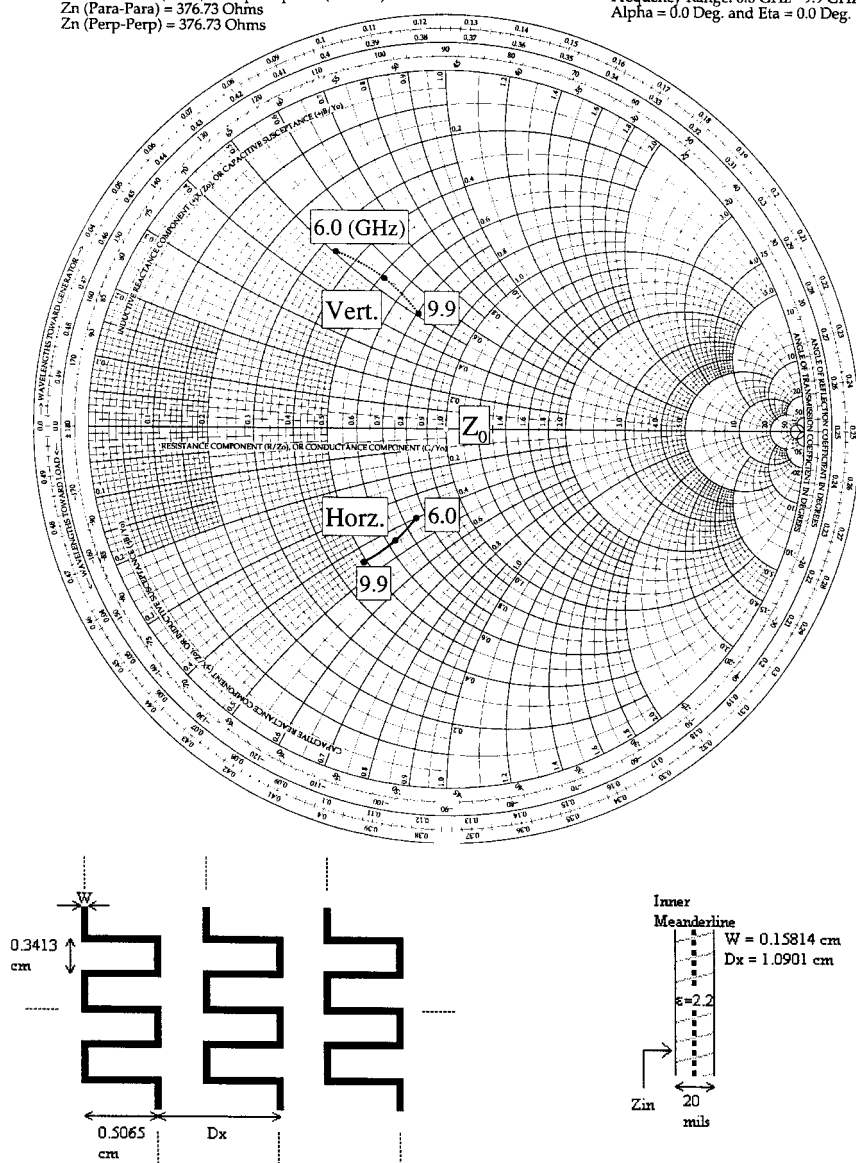


Fig. C.4 The input impedance of a single encapsulated meander line (inner) including free space Z_0 behind. Vertical as well as horizontal cases.

$f \sim 7.4$ GHz. At the top of Fig. C.6 we show the ratio between the vertical and horizontal components of the transmitted field as a function of frequency for various angles of incidence. Similarly, we show the phase difference between the two components in the middle of Fig. C.6.

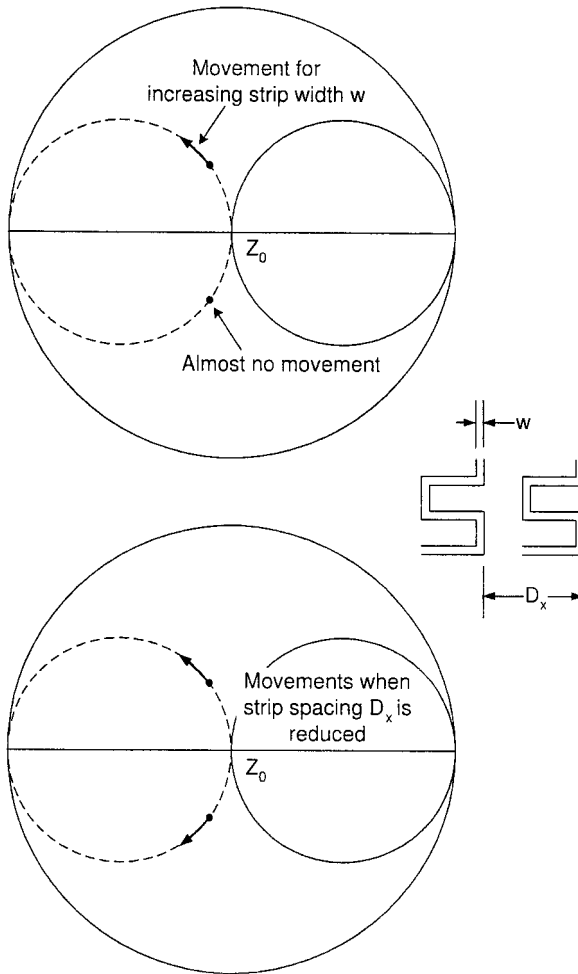


Fig. C.5 Typical movement of the meander-line impedance including free space Z_0 for: Top: Variation of line width w ; Bottom: Variation of line spacing D_x .

We observe a variation in amplitude ratio of less than 1.12 and a phase variation of less than 15° for normal angle of incidence. This corresponds to a cross polarization of less than ~ 17 dB.

However, at 60° angle of incidence we observe a serious deterioration of the amplitude ratio toward the lower frequencies, although it held up quite well at the higher frequencies.

The reason for this dilemma is simply that the electrical separation between the meander-line sheets typically is given by $\beta_0 d r_{0y}$ (d is the physical separation). Thus, at 60° the electrical separation will be reduced by a factor of $1/\cos 60^\circ = 2$, making the electrical separation approximately correct at the higher frequencies

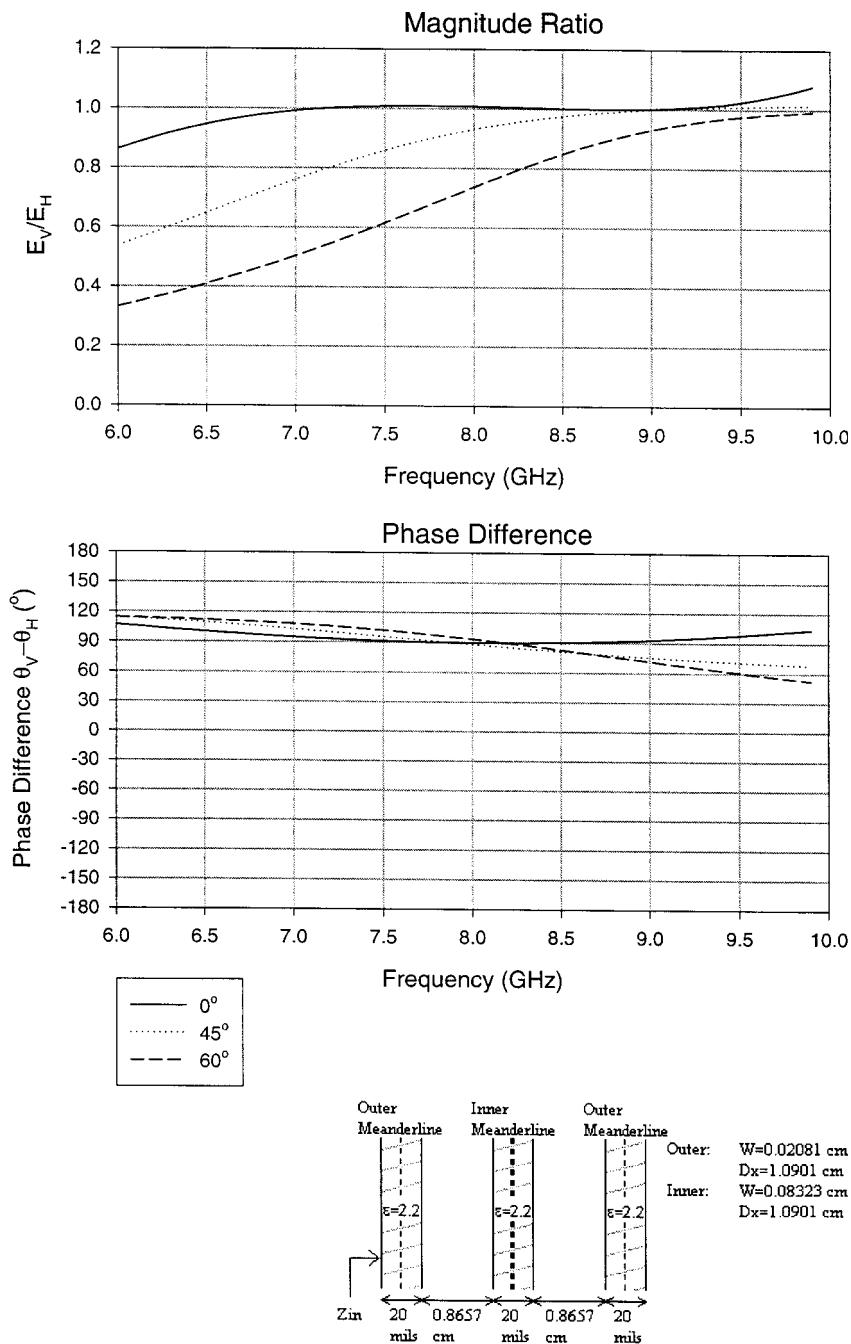


Fig. C.6 Design 1. Three encapsulated meander-line sheets without dielectric. Top: The magnitude ratio E_V/E_H as a function of frequency. Bottom: The phase difference $\theta_V - \theta_H$ as a function of frequency.

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 $\alpha = 0.0$ Deg. and $\eta = 0.0$ Deg.

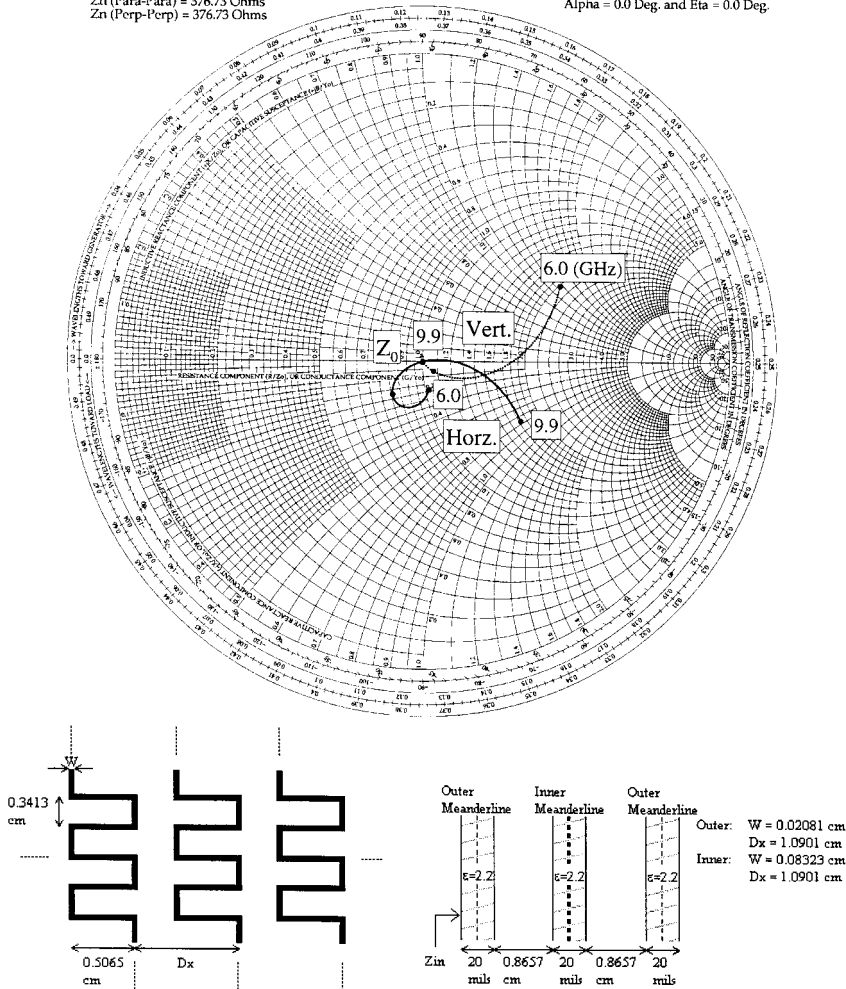


Fig. C.7 Design 1. Three encapsulated meander-line sheets without dielectric. Input impedances for the vertical as well as the horizontal field component.

but much too small at lower frequencies. Finally we show in Fig. C.7 the input impedances of the vertical and horizontal field components for the entire polarizer at normal angle of incidence.

C.5 DESIGN 2

Meander-Line Sheet Separation: $> \lambda_0/4$

Dielectric: Air

We just observed how Design 1 fails at the lower frequencies at higher angles of incidence because the factor r_{0y} becomes less than one. In Design 2 we shall attempt to correct for this discrepancy simply by increasing the meander-line sheet separation.

More specifically, the meander-line sheet separation is increased from 0.8657 cm to 0.9996 cm. It does indeed lead to some improvement in phase response at the lower frequencies but also to some degradation at the higher frequencies for normal angle of incidence. Thus, it is clear that a more radical treatment is needed. Since space is limited in this book, we have chosen simply not to show the results for this design.

C.6 DESIGN 3

Meander-Line Sheet Separation: $\sim \lambda/4$ in Dielectric

Dielectric Constant of Spacers: $\varepsilon_{r1} = \varepsilon_{r2} = 2.0$

In this design we will use dielectric slabs, rather than air, between the meander-line sheets as shown in Fig. C.8, bottom. The idea is simply that the electrical spacing in that case will be given by $\beta_2 d_2 r_{2y}$, where r_{2y} is the cosine to the angle of incidence inside the dielectric slab d_2 . Thus as a consequence of Snell's law r_{2y} will vary considerably less than r_{0y} in free space. In other words, We expect to obtain a more stable design with angle of incidence when using dielectric slabs instead of air. Indeed, the amplitude ratio shown in Fig. C.8, top, shows a significant improvement compared with nondielectric design 2, not shown. However, we also note that the phase difference shown in Fig. C.8, middle, is about 15° short of 90° , although it is very constant as a function of frequency.

The reason for this "shortage" of phase difference is that when we use dielectric spacers, we lower the intrinsic impedance by $\sqrt{\varepsilon_r}$. This simply implies that the meander-line impedances must also be lowered accordingly as will be demonstrated in the next design.

Finally, we show the input impedances for the vertical and horizontal field components in Fig. C.9. Note that the "center of gravity" is somewhat below Z_0 , which supports the statements made above.

C.7 DESIGN 4

Meander-Line Sheet Separation: $\sim \lambda/4$ in Dielectric

Dielectric Constant of Spacers: $\varepsilon_{r1} = \varepsilon_{r2} = 2.0$

Meander-Line Impedances Lowered

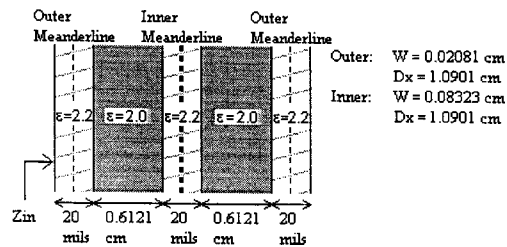
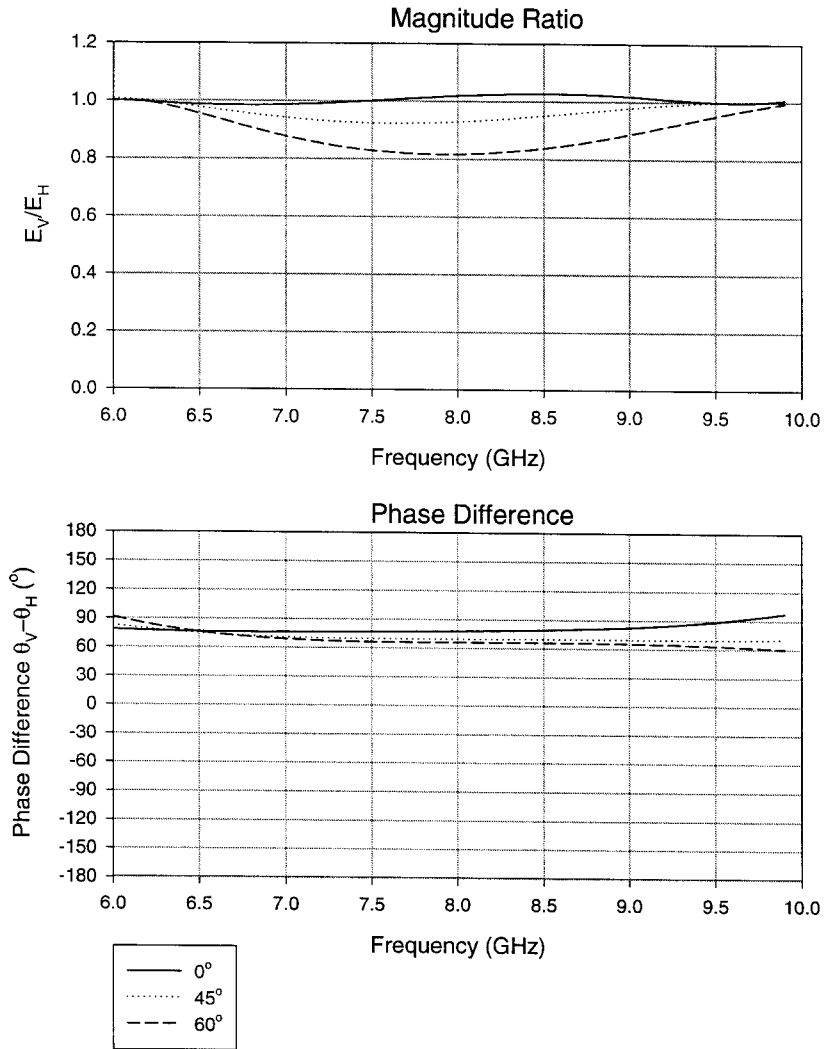


Fig. C.8 Design 3. Three encapsulated meander-line sheets with dielectric separation. Top: The magnitude ratio E_v/E_h as a function of frequency. Bottom: The phase difference $\theta_v - \theta_h$ as a function of frequency.

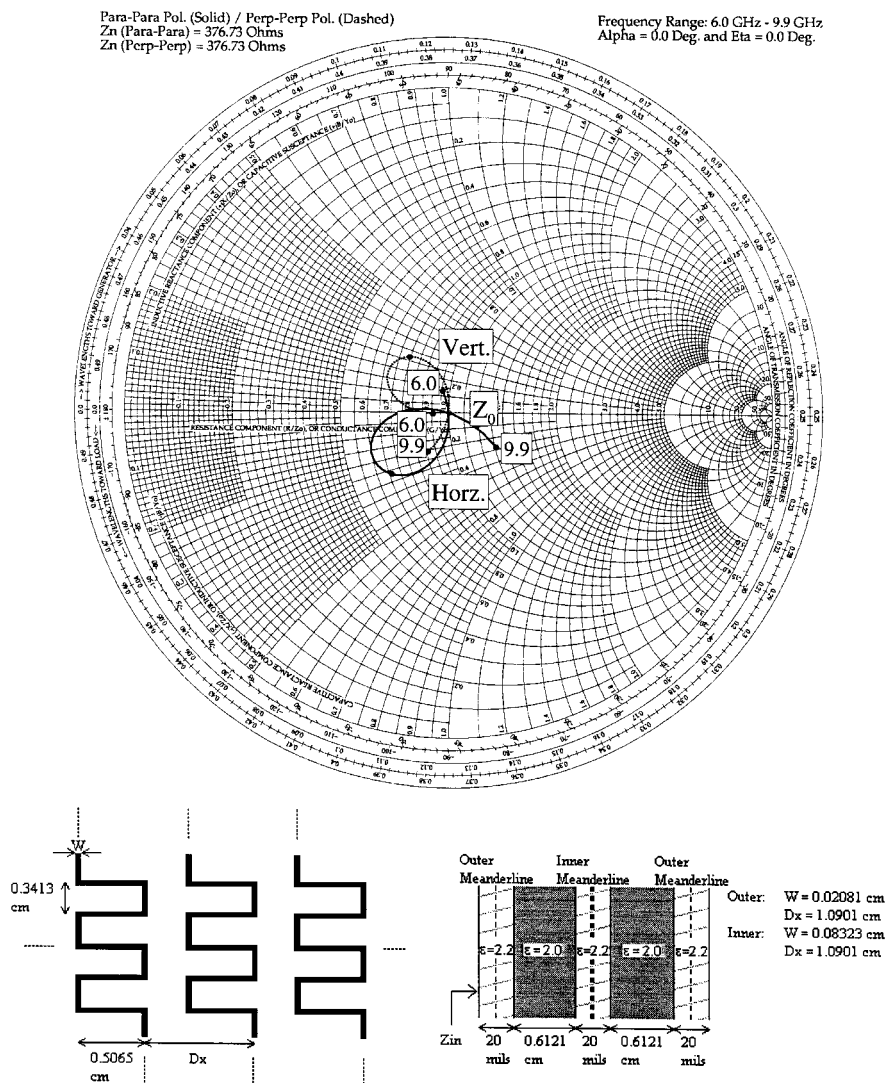


Fig. C.9 Design 3. Three encapsulated meander-line sheets with dielectric separation. Input impedances for the vertical as well as the horizontal field components.

This design has the same dielectric profile as Design 3, but the meander-line impedances have been reduced. This has been accomplished for the outer meander lines by simply increasing the line width w of the meander lines from 0.02081 cm to 0.03954 cm while all other dimensions remain the same. A plot of the outer meander-line sheet incorporating a dielectric slab is shown in Fig. C.10. It should be compared with the original meander-line sheet in Fig. C.3 except that the dielectric slab rotates the impedance approximately 90° clockwise (on the

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 $\alpha = 0.0$ Deg. and $\epsilon = 0.0$ Deg.

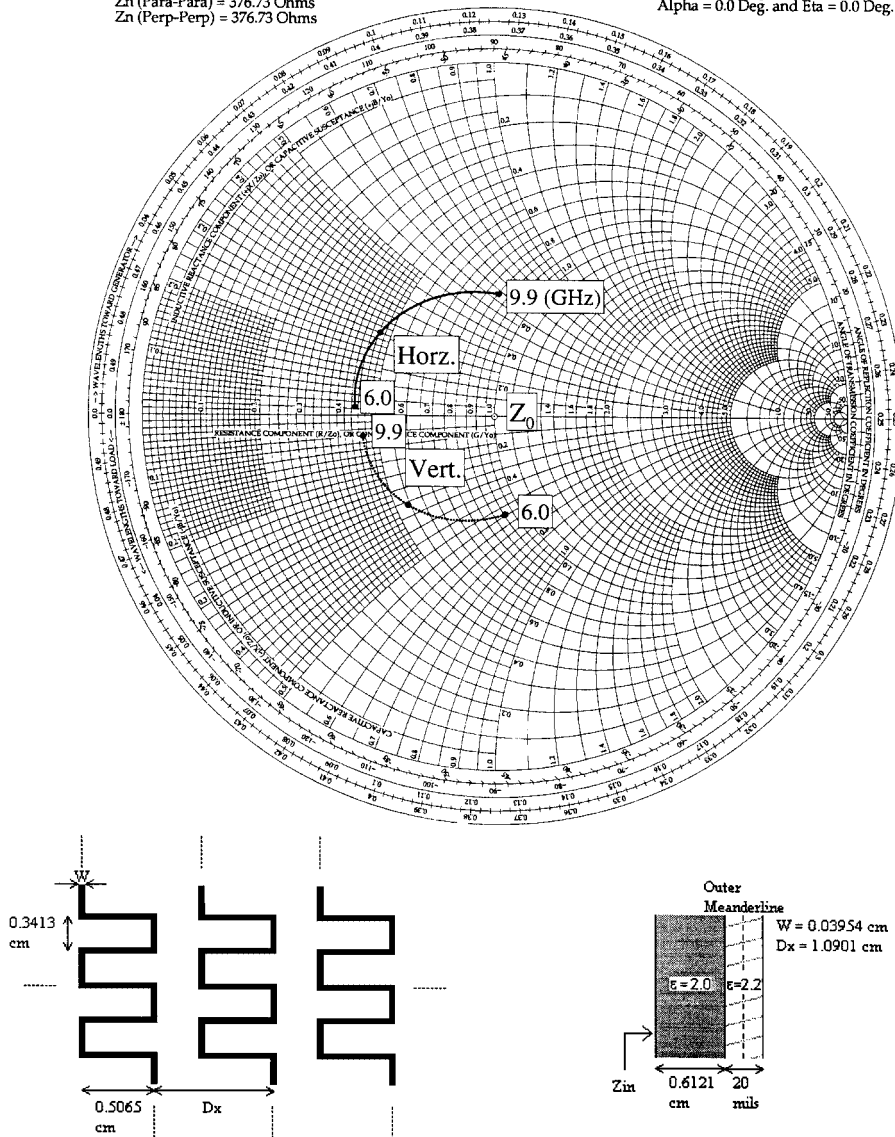


Fig. C.10 Design 4. Input impedances (vertical and horizontal) of a single meander line (outer) including free space Z_0 behind and a dielectric slab in front.

average). Similarly, the impedance of the inner meander-line has been lowered by increasing its line width from 0.08323 cm to 0.15814 cm while all other dimensions remain the same. The input impedances of the outer and an inner meander line are shown in Fig. C.11, while the addition of the next dielectric slab is

Para-Para Pol. (Solid) / Perp-Perp Pol. (Dashed)
 Zn (Para-Para) = 376.73 Ohms
 Zn (Perp-Perp) = 376.73 Ohms

Frequency Range: 6.0 GHz - 9.9 GHz
 Alpha = 0.0 Deg. and Eta = 0.0 Deg.

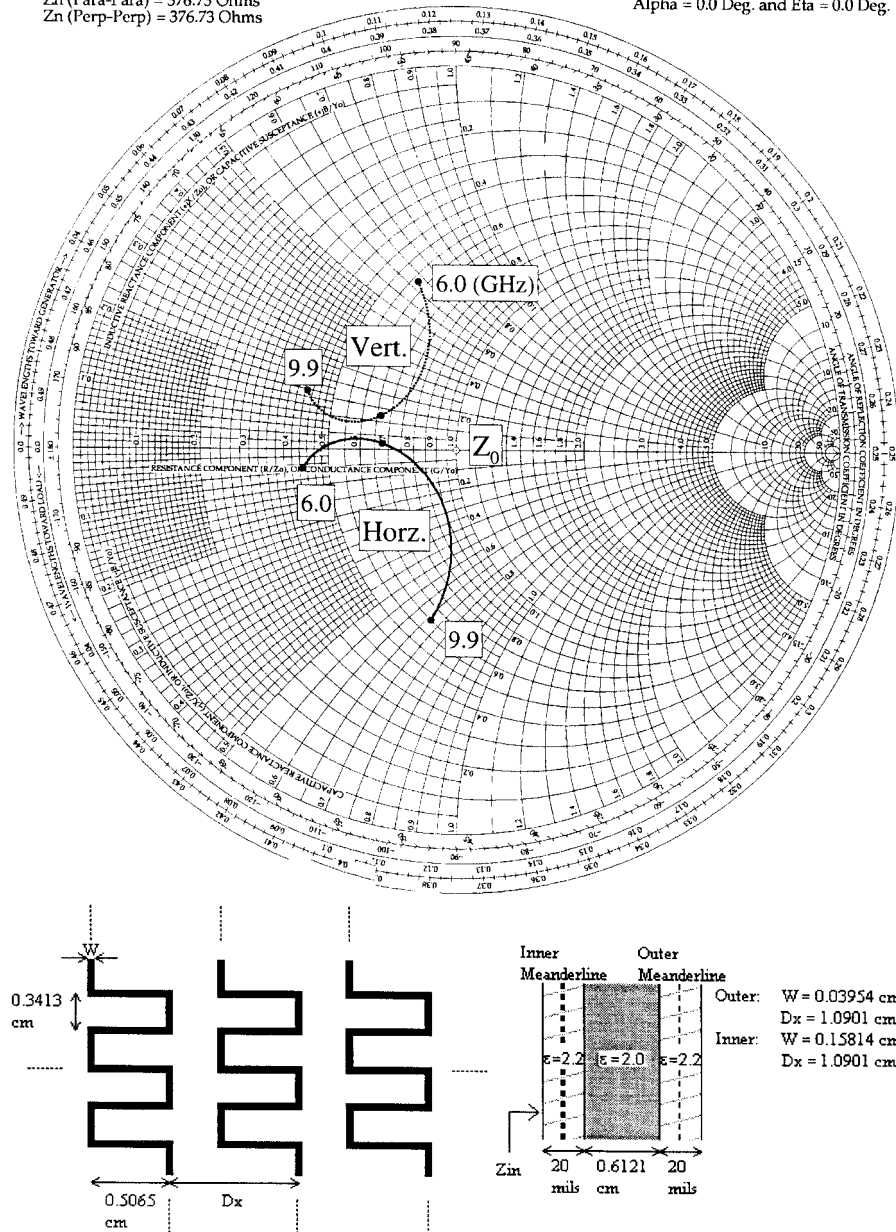


Fig. C.11 Design 4. Input impedances (vertical and horizontal) of two meander lines separated by a dielectric slab.

Para-Para Pol. (Solid) / Perp-Perp Pol. (Dashed)
 Z_0 (Para-Para) = 376.73 Ohms
 Z_0 (Perp-Perp) = 376.73 Ohms

Frequency Range: 6.0 GHz - 9.9 GHz
 $\alpha = 0.0$ Deg. and $\eta = 0.0$ Deg.

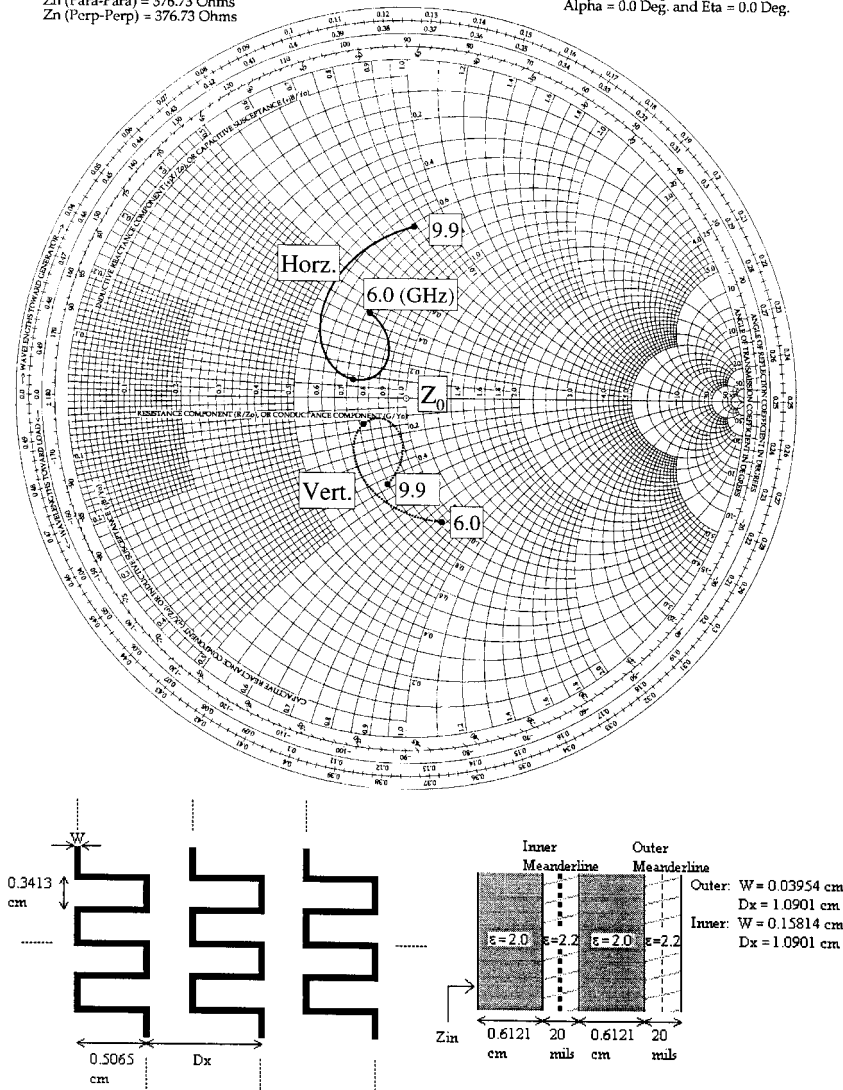


Fig. C.12 Design 4. Input impedances (vertical and horizontal) for two cascaded meander-line sheets and two dielectric slabs as shown.

shown in Fig. C.12. Finally, in Fig. C.13 we have added the last outer meander line. Note here the input impedance remains close to Z_0 over a considerable bandwidth. Thus, it should not surprise us that the amplitude ratio as well as the phase difference shown in Fig. C.14 looks almost perfect for normal angle of incidence and “not bad” for oblique angle of incidence.

However, further improvement is possible, as we shall see next.

Para-Para Pol. (Solid) / Perp-Perp Pol. (Dashed)
 Z_n (Para-Para) = 376.73 Ohms
 Z_n (Perp-Perp) = 376.73 Ohms

Frequency Range: 6.0 GHz - 9.9 GHz
 $\alpha = 0.0$ Deg. and $\epsilon = 0.0$ Deg.

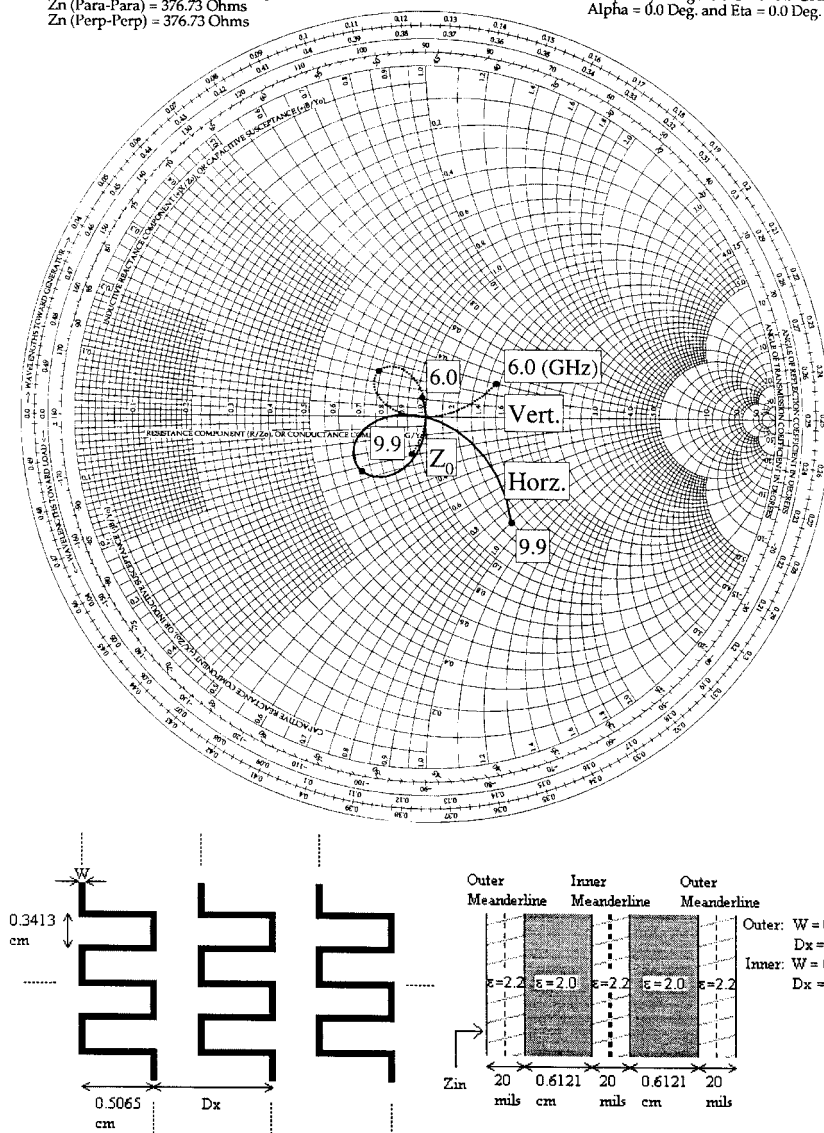


Fig. C.13 Design 4. Input impedances (vertical and horizontal) for three cascaded meander-line sheets separated by dielectric spacers as shown.

C.8 DESIGN 5

Meander-Line Sheet Separation: $\sim \lambda/4$ in Dielectric

Dielectric Constant of Spacers: $\epsilon_{r1} = \epsilon_{r2} = 2.0$

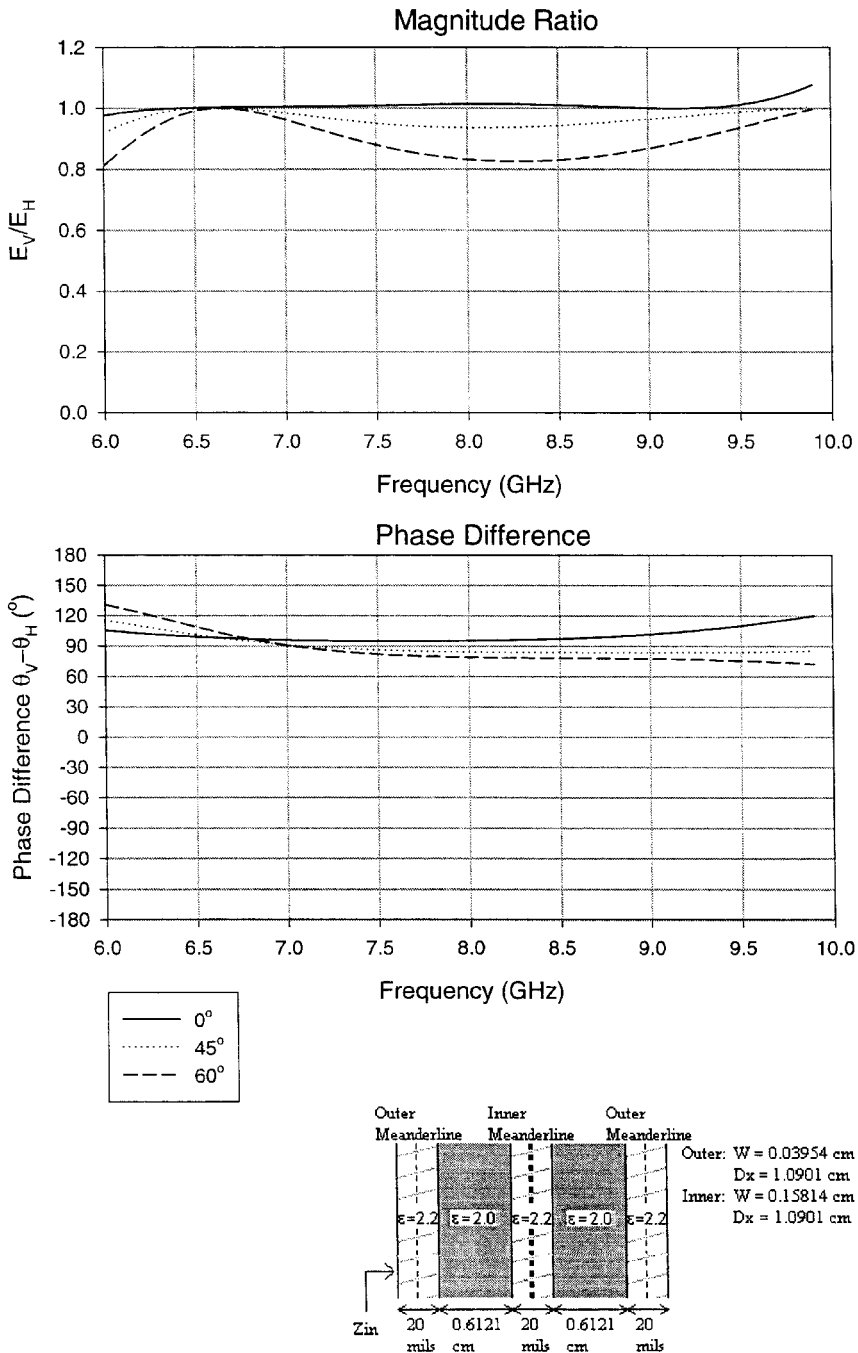


Fig. C.14 Three meander-line sheets separated by dielectric slabs. Top: The amplitude ratio E_V/E_H as a function of frequency. Bottom: The phase difference $\theta_V - \theta_H$ as a function of frequency.

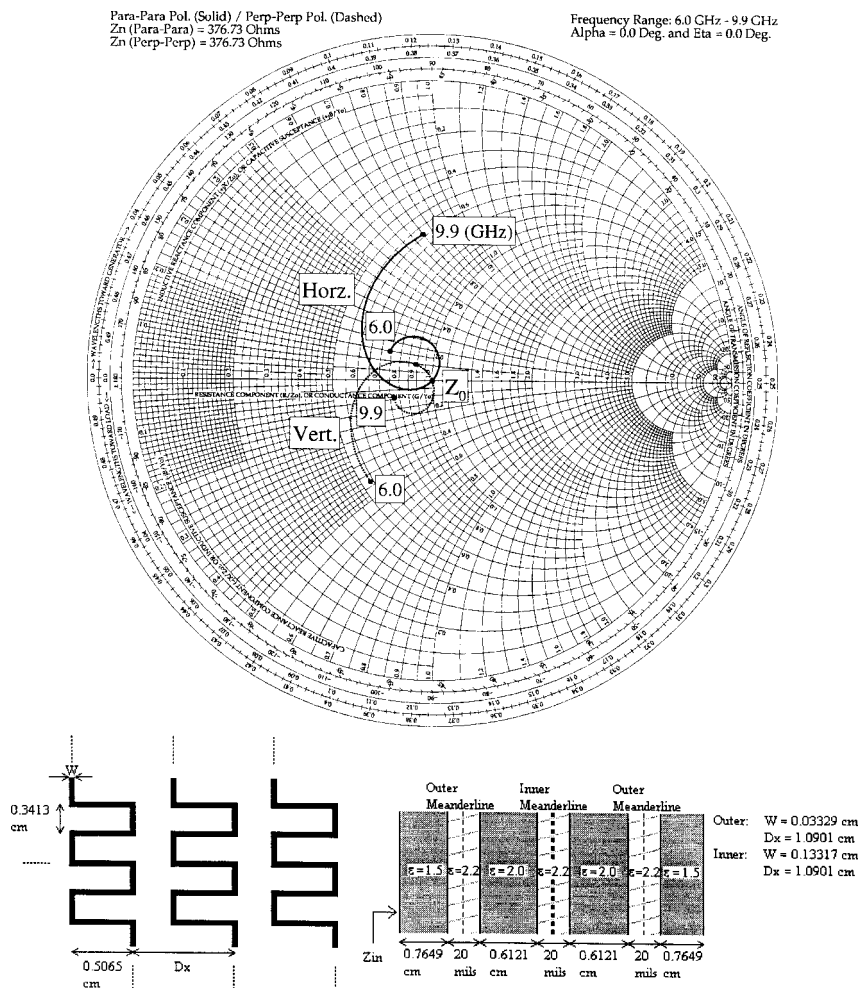


Fig. C.15 Design 5. Three meander-line sheets separated by dielectric slabs and further provided with outer matching transformers. The input impedances (vertical and horizontal).

Meander-Line Impedances Lowered Outer Matching Plate at Each End

It is by now well known that the variation with angle of incidence of the scan impedance of phased arrays as well as the bandwidth of hybrid radomes can be reduced by using dielectric slabs placed between free space and the device in question. To be sure, the dielectric constant should in general be less than 2 (for a single slab) and the thickness should be somewhat thicker than $\lambda/4$ in the dielectric. An example of applying this technique is shown in Fig. C.15. Compared to the uncompensated case in Fig. C.13, we observe some improvement

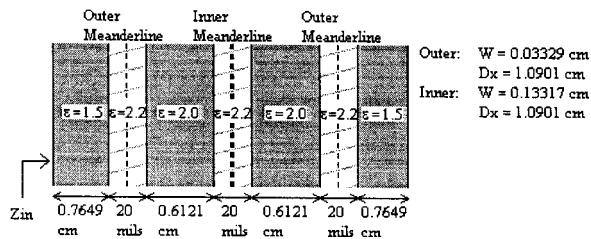
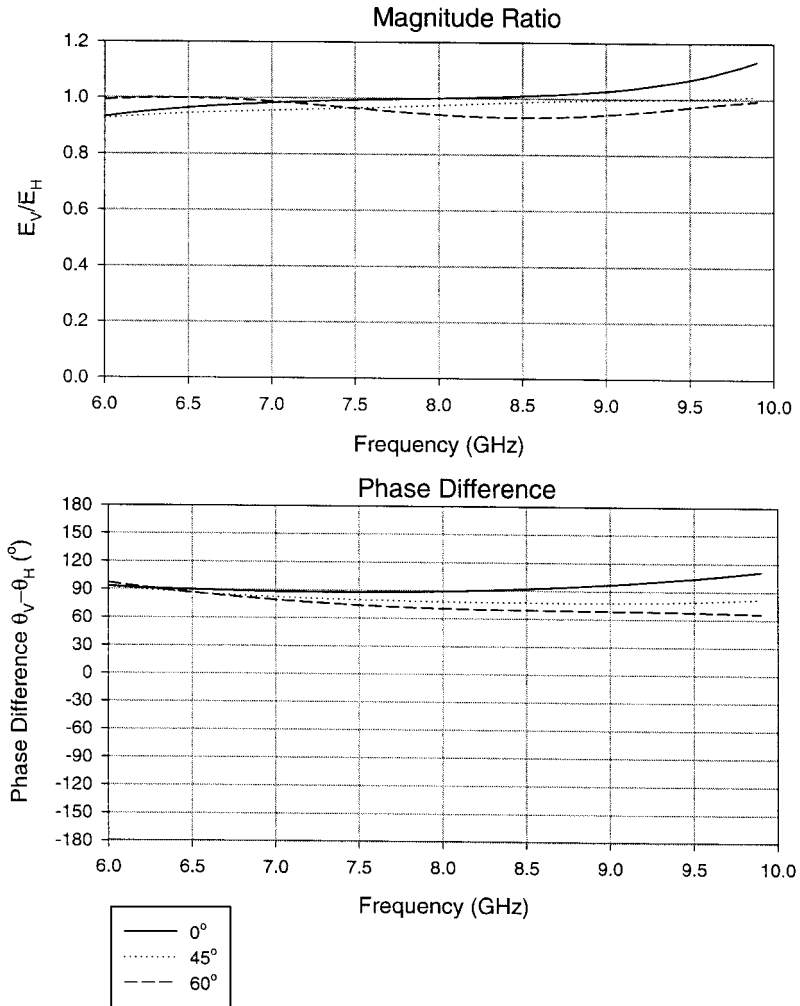


Fig. C.16 Design 5. Three meander-line sheets separated by dielectric slabs and further provided with outer matching transformers for wide angle scanning. Top: The amplitude ratio E_V/E_H as a function of frequency. Bottom: The phase difference $\theta_V - \theta_H$ as a function of frequency.

for horizontal polarization at the lower frequencies and for vertical at the higher frequencies. Note, however, we only show normal angle of incidence.

Furthermore, we show in Fig. C.16 the amplitude ratio as well as the phase difference as a function of frequency. Compared to the uncompensated Design 4 shown in Fig. C.14, we observe a significant improvement in amplitude ratio, less so in the phase difference. It should be pointed out, however, that neither Design 4 nor Design 5 has been optimized.

C.9 CONCLUSION

Examination of Design 6 comprised of four meander-line sheets with dielectric spacers both inside and outside showed that some improvement compared to Design 5 was possible. However, since the differences between these specific designs were minor, it was decided not to show Design 6. It is the opinion of the author as well as his right-hand man, Jonothan Pryor, that further improvement is possible by a simple optimization process (which algorithm is used is of less importance). We shall leave this as an exercise for the student.

PROBLEMS

C.1 Consider a polarizer comprised of two meander-line sheets.

Design it to yield perfect circular transmission at the center frequency. Determine the reflection coefficient at that frequency.

What percentage of the incident field is being transmitted and how much is being reflected?

Compare it to a polarizer of a single meander-line sheet.